

let a, b be coprime, $a < b$

Aside:

$$10 \equiv 3 \pmod{7}$$

Conjecture: Any number n can be repeatedly added under modulo n s.t. the value is $\neq 0$
1 or $n-1$.

if $k \in [2, n-2]$:

$$\begin{aligned} & a^k \pmod{n} \\ &= \left(\left((k \cdot k) \pmod{n} \cdot k \right) \pmod{n} \cdot k \right) \pmod{n} \end{aligned}$$

if k is even this conjecture is proved

if k is odd:

$\because a, b$ are coprime, and $a < b$

$$b \equiv k \pmod{a}$$

where k is coprime with a

If not coprime $b-a = f \cdot n_0$, $f < a$
and $a \equiv 0 \pmod{f}$, and $b \neq Na$,

If $b-a \leq \frac{a}{2} \Rightarrow b \leq 3 \times \frac{a}{2}$
and a, b are not coprime, then

$$b \equiv k \pmod{a}$$

$\Rightarrow k$ is $\gcd(a, b)$ ^{e.g.} ~~except $\gcd(a, b) = a$~~
 _{$\because b \in Na$}

If $b \geq \frac{3}{2}a$

$$b \equiv k \pmod{a}$$

$$\begin{array}{r} 45, 36 \\ 12, 21 \end{array}$$

$$\Rightarrow K = \gcd(a, b) \cdot N,$$

$\therefore$ if 2 numbers are not coprime and $a < b$
 $a \equiv b \pmod{a}$ is not a relation then

$$K \equiv b \pmod{a} \text{ has } k = \gcd(a, b)$$

e.g. $16, 56 \mid 56-48=8 \checkmark$

e.g. $7, 9 \mid 9-7=2 \times$ but coprime $\checkmark$

$$14, 25$$

$\therefore$ if $a < b$, a, b coprime then

$$b \equiv k \pmod{a} \text{ } k \text{ coprime with } a$$

if k is not coprime with (a, b) , it's a factor

Question

pre

$$5, 7 \quad k=2$$

$$\begin{array}{l} 20, 21 \\ 4, 5, 3, 12 \end{array} \quad 15, 21 \quad 15, 14$$

coprimality is not transitive

$$7, 10 \quad k=3$$

$$21, 30 \quad k=2$$

$$7, 19 \quad k=5$$

$$\begin{array}{l} 56, 57 \\ 19, 58, 59 \\ 5, 10, 15 \\ 5, 3, 1 \end{array}$$

for a, b coprime, $a < b$,
for $b \equiv k \pmod{a}$ k is coprime with (a, b)
by contradiction. —

Does $\langle k \rangle_{\text{mod } a}$ generate $\{1, a-1\}$?

Yes the group generated by k under mod a is

$\langle k \rangle_{\text{mod } a}$ where k is coprime with

a generates the cyclic group under mod a
thereby containing 1 or $a-1$. By intuition and approx. $\phi(a)$

$$\therefore \gcd(a, b) = au + bv$$

$$\Rightarrow 1 = au + bv \text{ for some } u, v \in \mathbb{Z}$$

for a, b coprime

Proof: $b - a$

$$= \gcd(a, b) [B - A]$$

B, A are coprime

$$\therefore \exists \text{ an } n \text{ s.t. } n \cdot k \equiv 1 \pmod{a}$$

$$\text{or } nk \equiv a-1 \pmod{a},$$

$$\text{where } k \equiv b \pmod{a}$$

$$\therefore \exists \text{ terms } u, v \in \mathbb{Z} \text{ s.t.}$$

$$1 = \gcd(A, B) = Au + Bv$$

$$\Rightarrow b - a$$

$$= \gcd(a, b) [B - A]$$

$$= \gcd(a, b) (Au + Bv) [B - A]$$

$$\Rightarrow \frac{b-a}{[Au + Bv](B-A)} = \gcd(a, b)$$

$$\Rightarrow \frac{b}{B-A} - \frac{a}{B-A} = \gcd(a, b)$$

$$\Rightarrow \therefore u, v = B - A$$

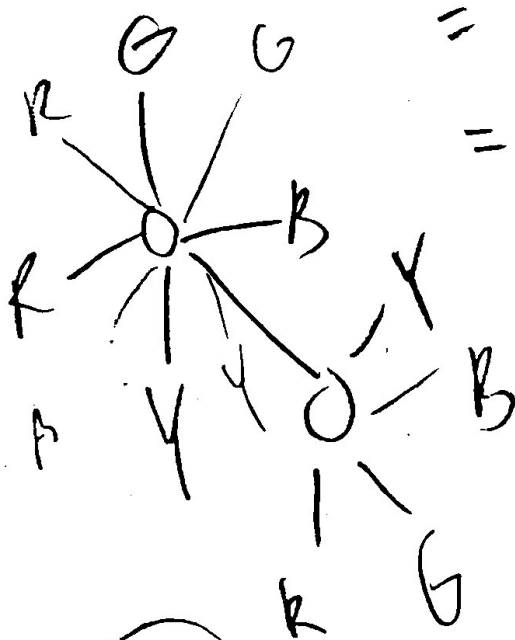
Proved!!!

$$\Rightarrow \frac{b-a}{B-A} = \gcd(a, b)$$

$$4n+2$$

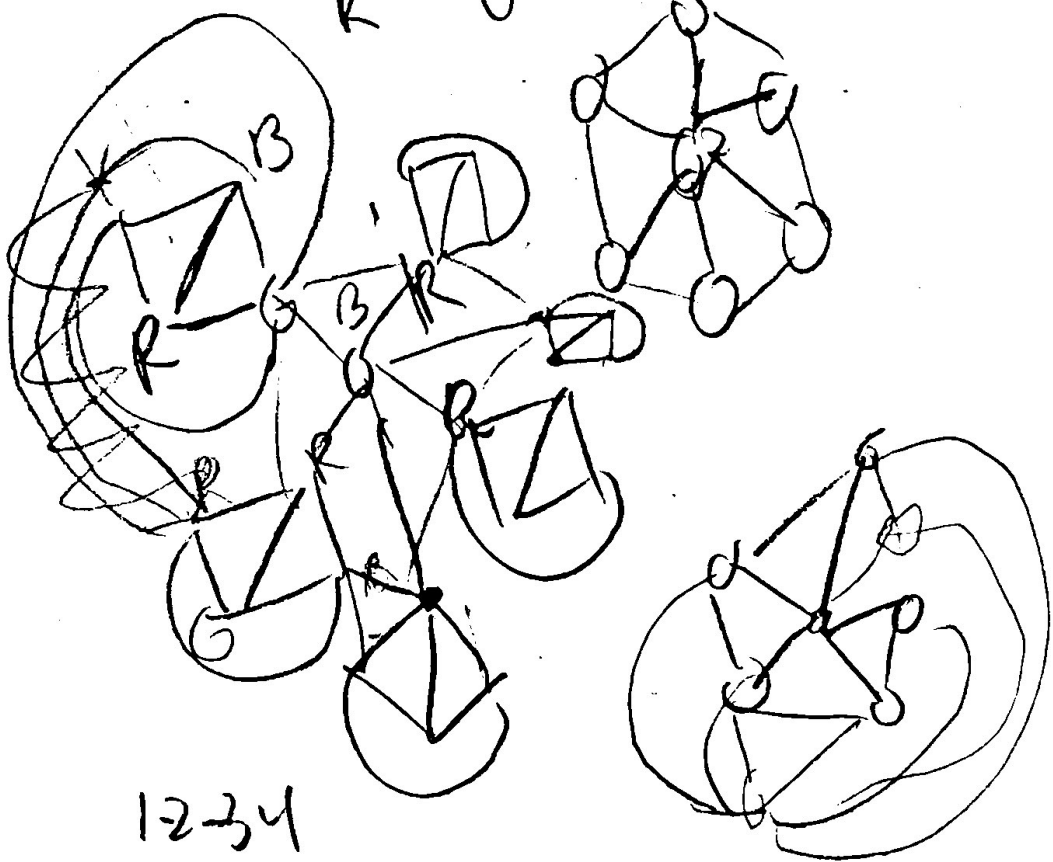
$$= 2(2n+1)$$

$$= 2.$$



$$1.1 + 2$$

$$1.2 + 2$$



1234

12046

1230458

1. Naïve Just color.
- may not work in the end

2. Naïve Just color, but edges
may not work at all

3.

How to make graph maximally planar?

planar $\Rightarrow$ no K_5 or $K_{3,3}$

$$e = 3v - 6$$

for maximally planar $|V| \geq e - v + 2$

$$\sum_{i=1}^R \deg(v_i) = 3r$$

$$3r = \sum_{i=1}^V \deg(v_i)$$

$$= 2e$$

$$3(e - v + 2) = 2e$$

$$3e - 3v + 6 = 2e$$

$$e = 3v - 6$$

$$\frac{e}{v} = \frac{3v - 6}{v} = 3 - \frac{6}{v}$$

$$\frac{e}{v} = \frac{3v - 6}{v}$$
$$\frac{e}{v} = 3 - \frac{6}{v}$$
$$\text{avg} = 3 - \frac{\sum \deg(v_i)}{v}$$

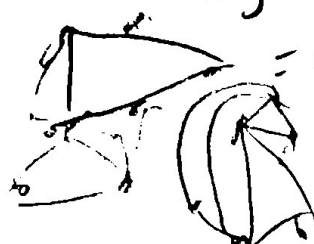


$$= 2e$$

$$= 6v - 12$$

$$\deg(v_i) \text{ avg}$$

$$= 6 - \frac{12}{v}$$



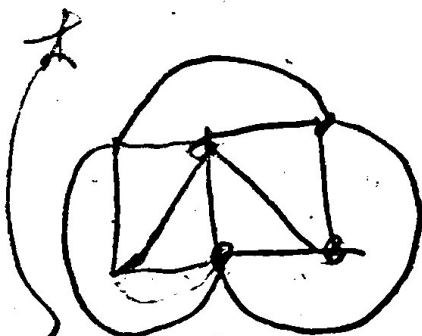
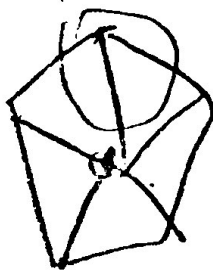
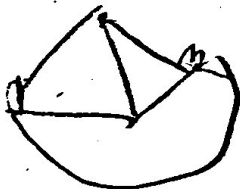
$$e = 3v - 6$$

$$= 3(v-1) - 6$$

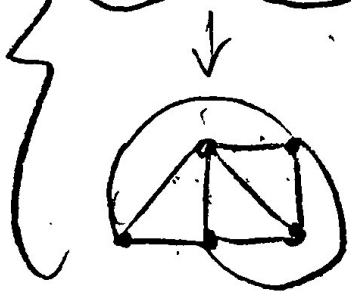
$$= 3v - 3 - 6$$

$$= 3v - 9$$

$$= e_1 - 3$$



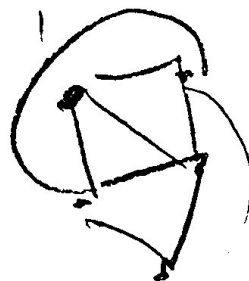
$$3(6) - 6 = 12 \text{ edges}$$



key: 4

9 edge

3v-6

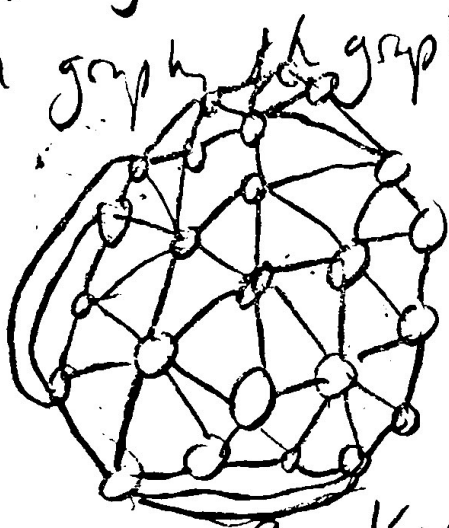


Theory: The set of uncoloured nodes and their boundary of coloured nodes is

maximal

(*) Laprock

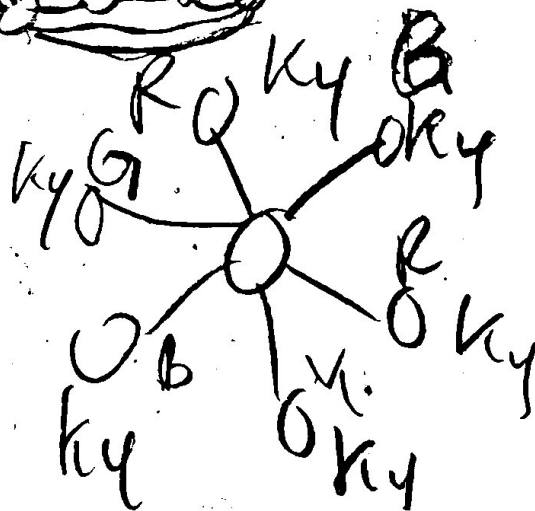
Theory: by colouring all nodes with ≤ 6 neighbours on a graph the graph will be coloured



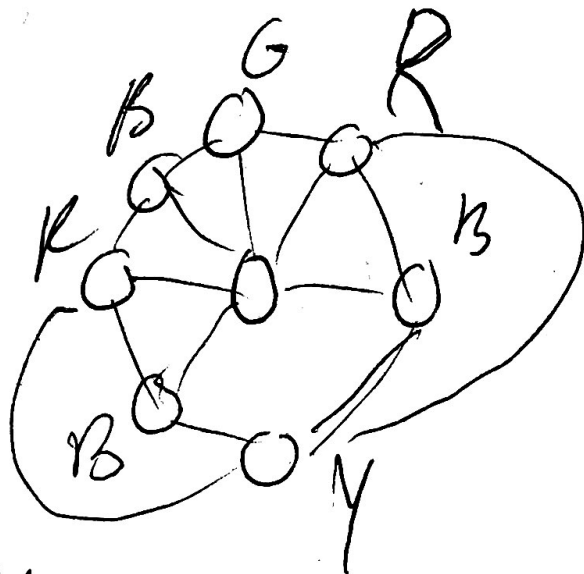
ordering:

uncoloured adjacent neighbours

in this order

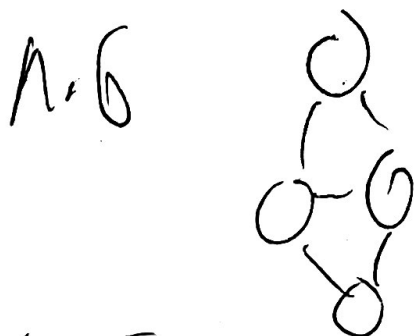


Generating maximally planar graph is not easy



- algorithm's complexity
- finish algorithm
- notes on optimizations
- test cases, edge case,
- corner cases

average number of vertices less than 6.

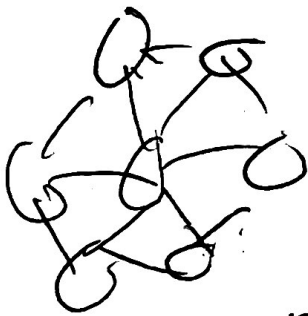
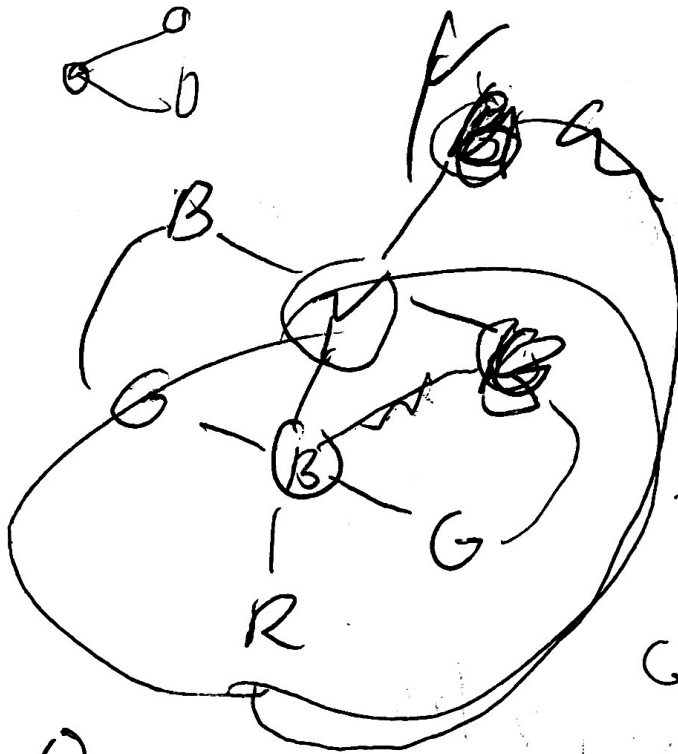


$$n = 5 \cdot n = 5n^2$$

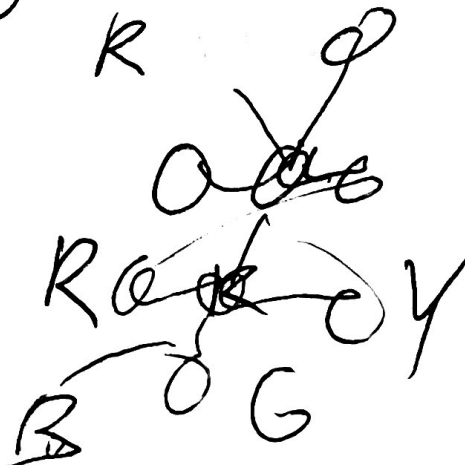
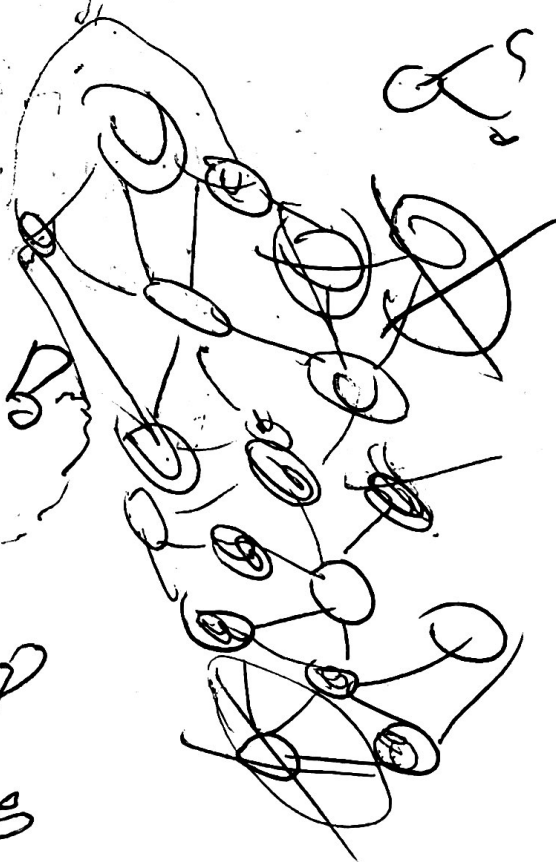
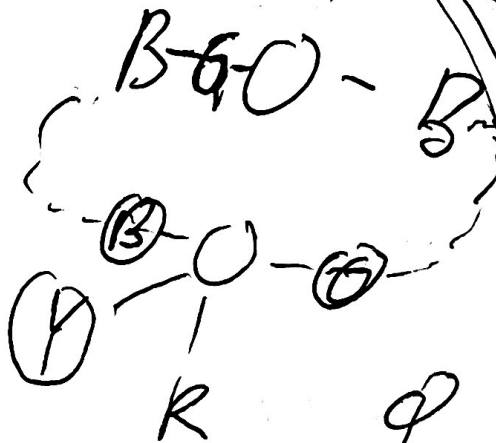
$$n = 5n$$

$$5 \cdot 5n^3 = 25n^3 = O(n^3)$$

Question hub
color neighbors?



mn



key does not have any suffix,
= no tape chunk

if uncovered $\Rightarrow$ no connection
no edge

for node:

go to neighbors

if measurable try to connect

using RGS

if fail keep it

it is edge next...

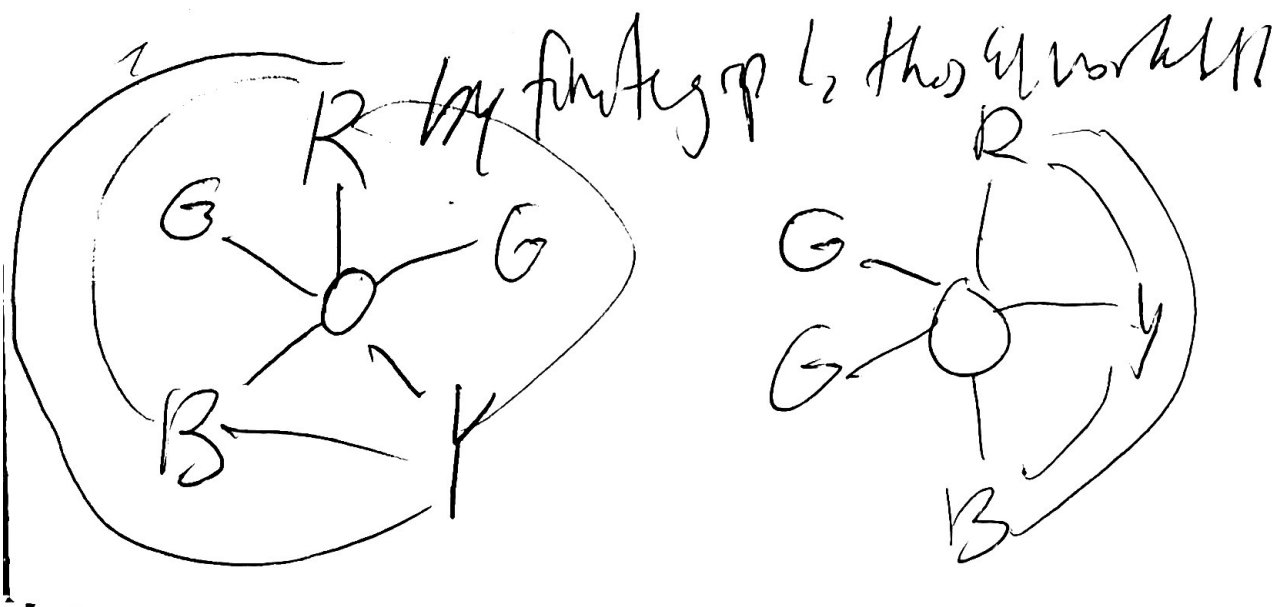
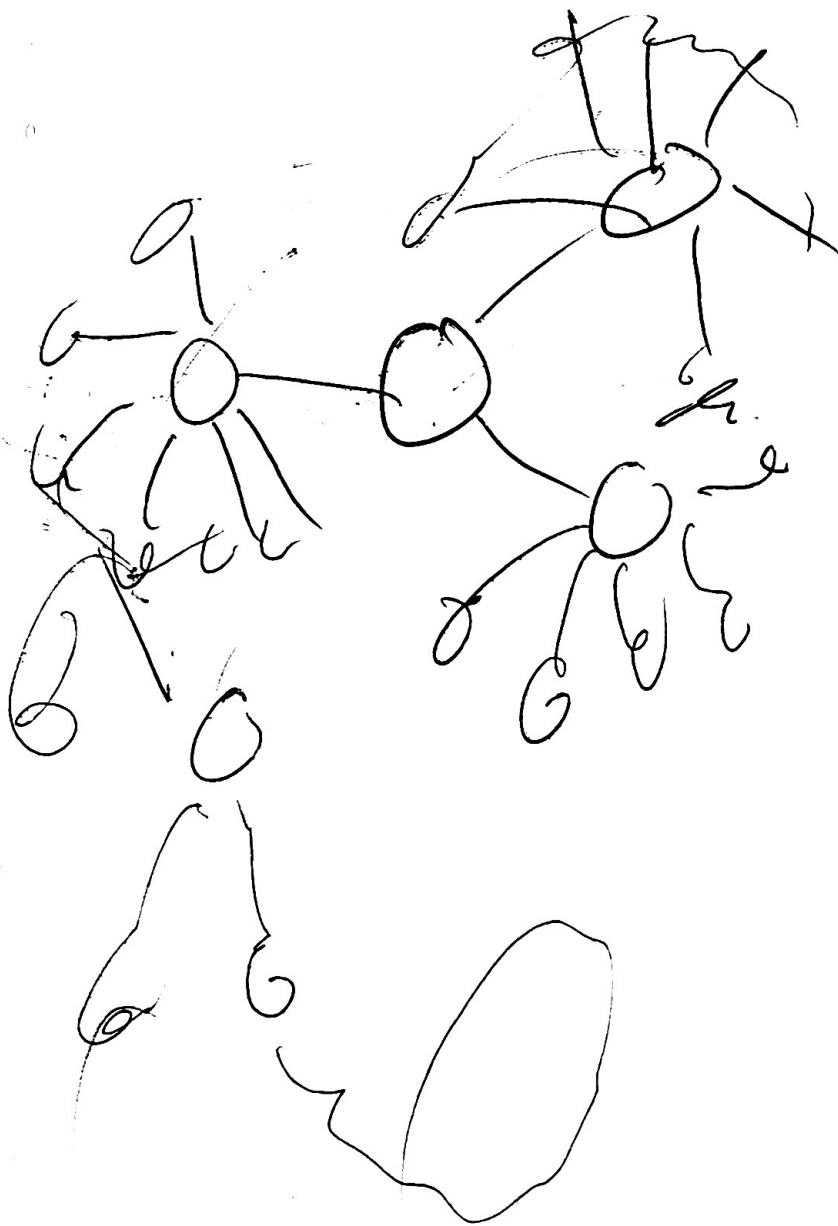
~~then edge node~~

for a node: go to neighbors and try to
cover each neighbor

- if discover w/ ≤ 43 , don't

- if not, use keep's method w/ measuring
node n as an wedge

- if $\leq 5 > 6$ and measurable go next,
by finite res, it should work out



removing all colored nodes w/ colored
neighbors $\Rightarrow$ no Kempe chains

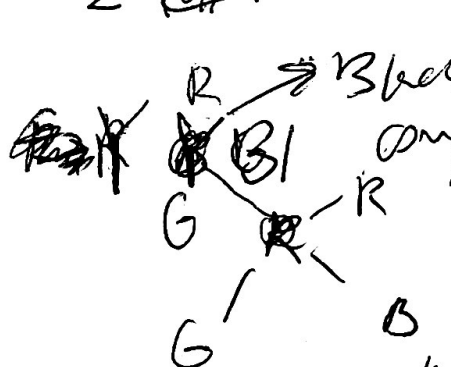
1st idea:

- removing colored nodes can destroy
Kempe chains.

o rectcheck and use SCT if destroyed.

2nd idea:

- 2 ~~far~~ removed nodes can not error

 B has 0 A selected node has
complete

o so of neighbors, so the
neighbor never get selected

It can't get selected since it is already
colored.

3rd idea:

- colouring the neighbors can fail.

Assume ~~greater~~ greater than 6 colored nodes on
each neighbour of a uncolored node of
 $\delta < 6$.

The subgraph

* Works for all nodes w/ ≤ 6 color
neighbours and colour.

Assume all nodes have $\neq$ neighbours

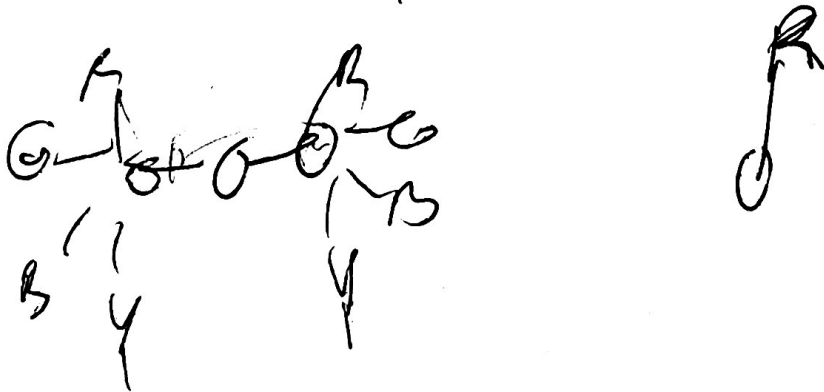
If uncoloured originally, greedy nodes will
work

If coloured properly guarantee will
fail If coloured properly can still fail

→ indicate by graph has been code
outputs

our graphs should have no longer than

→ success always, otherwise, not possible



key facts

- colours inputted in must be allowed the same way, otherwise an ~~non~~ uncoloured graph is input

$\Rightarrow$ colours inputted in can not be $\{R, G, B, Y\}$ around an uncoloured node.

- Just don't consider - originally coloured nodes

- it doesn't make sense, they're fixed

but does algorithm accommodate?

accommodate if/then

say unresolvable using VET.

SCT is after C.

We prove SCT:

for if colouring we show it possible w/ given

Cases:

① uncoloured graph V

② coloured graph but can change colours originally

SCT. $\Rightarrow$ $V \cup$ Just an uncoloured graph

③ coloured graph but can't change colours

- OWS, have to mark when SCT fails for a specific graph w/ predefined colouring

$\Rightarrow$ SCT always works by induction on uncoloured graph case 0, 1.

(3) fails when all uncoloured nodes share ~~exactly~~ neighbours $\delta \geq 5$ w/ all δ colours "failing condition"

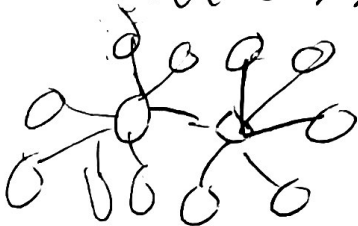
Algorithm

never assume an originally coloured node is unchangeable, try to colour the graph by SCT.

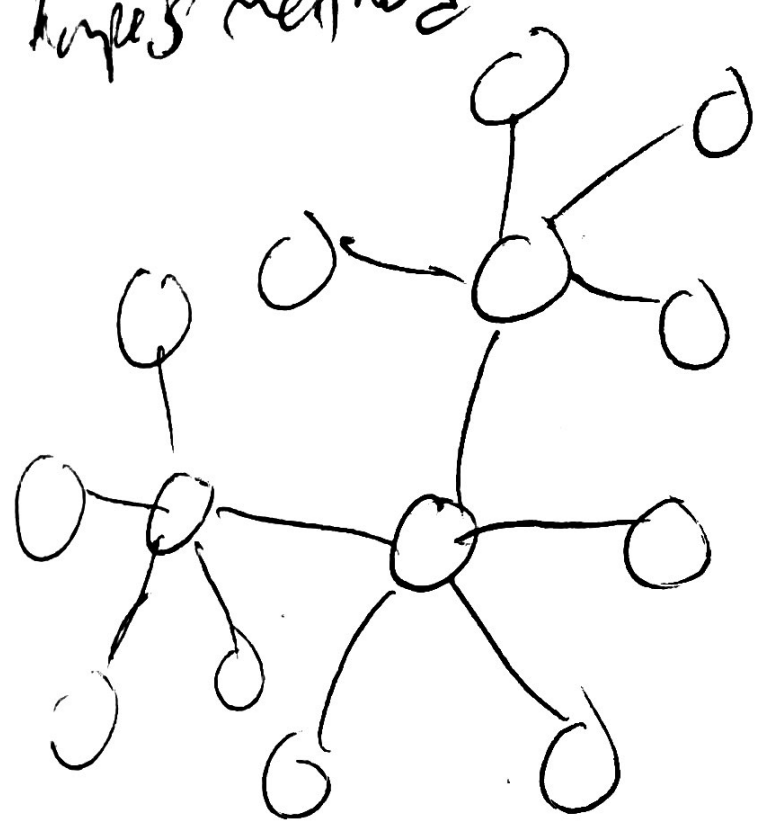
If fails $\Rightarrow$ failing condition & break
- The failing condition is irrelevant since we just want to prove

Kempe's method $\Rightarrow$ SCT.

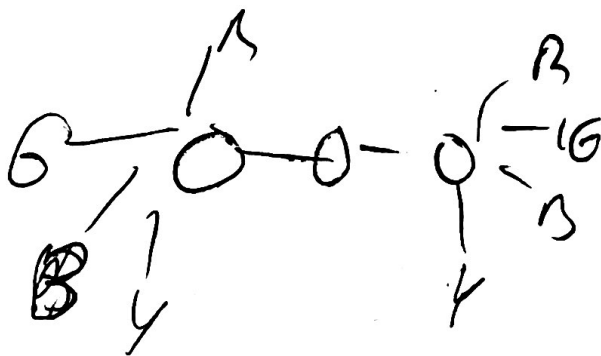
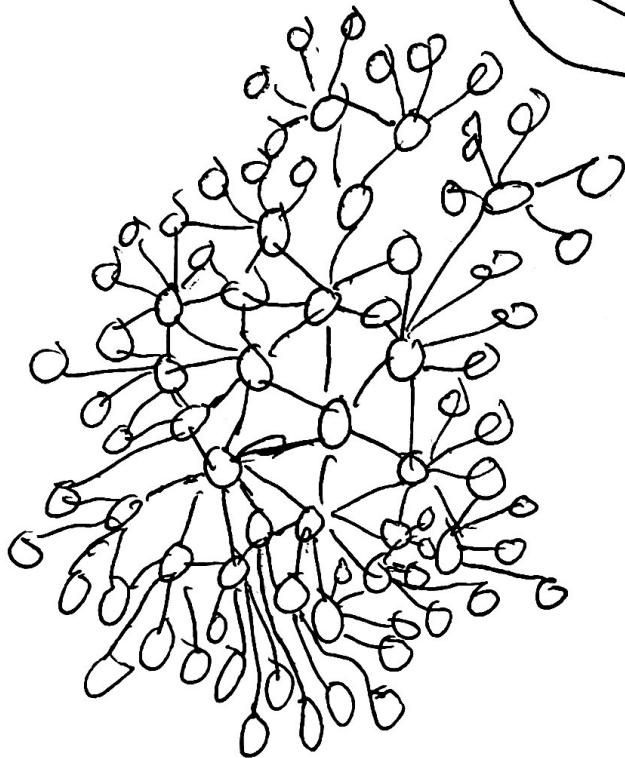
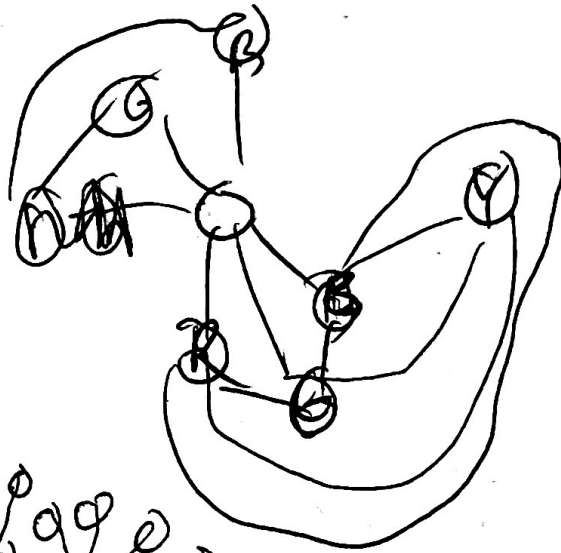
prove SCT for counterexample s.t. all cases of Kempe's method works.

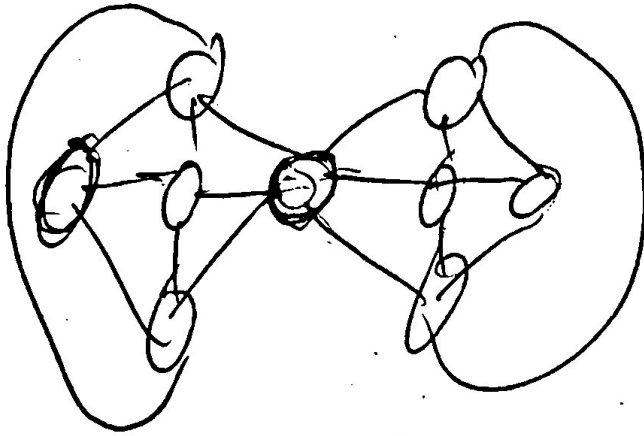


huffman method



Question 1





new nodes, ⁱⁿ nodes on uncoloured nodes,
 coloured nodes, required to be
 4 coloured, by induction.

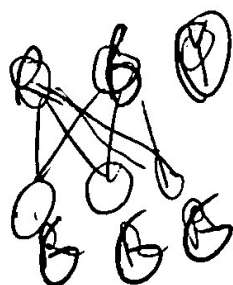
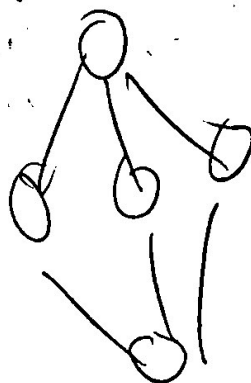
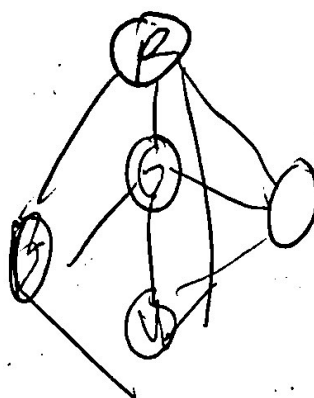
Implement restore function

restore:

node n

find n in copy,

If any neighbours are in graph, rebuild
 it. within, otherwise



if n_k minor works

Does Karp's method prove $t=6$ for
Hindman's conjecture?

$t=6$
no K_6 minor $\Rightarrow$ 5-colorable

• ~~no~~ K_5

$e \leq 3v-6$ for planar graph

No K_6 minor proof is 5-colorable by
induction using Karp's method

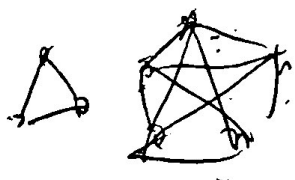
assume no K_5 minor ^{planar} $\Rightarrow$ 5-colorable

assume has K_5 minor

^{edges} $3v-6$ for maximally planar graph

add an edge to every vertex

$4v-6$



$$3^C 2 = \frac{3!}{2!}$$

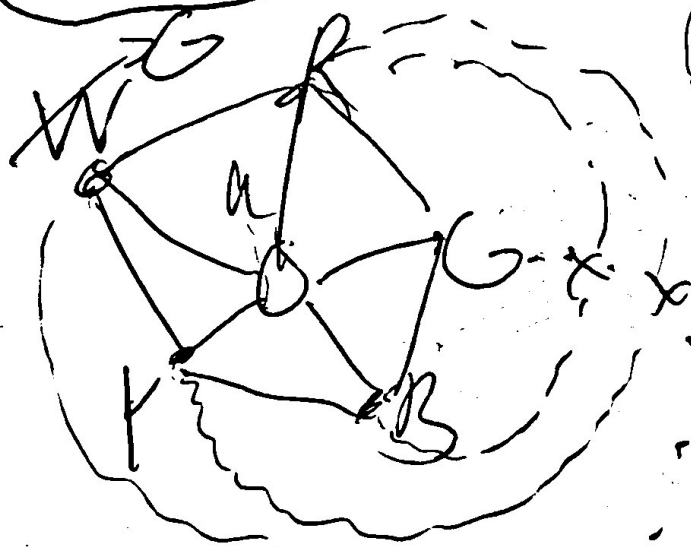
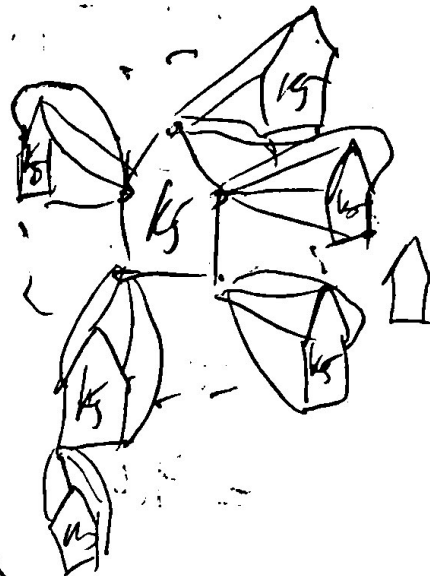
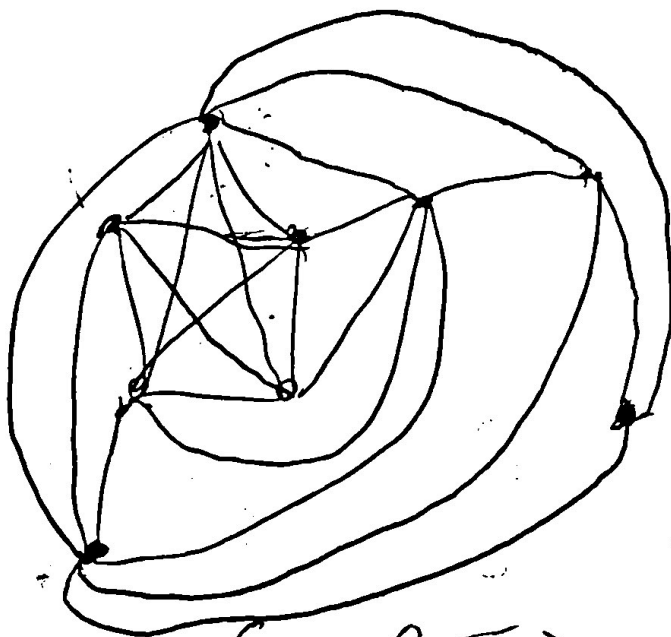
$\leq 3v-6$

$v^C 5$

$$\frac{v!}{(v-5)!}$$

$3v-6$
 $6v-12$

$$+ v^C 5 + \frac{v(v-1)(v-2)(v-3)(v-4)}{5!} \leq 3v-6 + v^C 5$$



$$\frac{n(n-1)}{2} = \frac{n \cdot 4}{2} = 2n \text{ edges}$$

maximally $k_5 \Rightarrow$ non planar

$$[n-4] k_{55} \Rightarrow 10 \text{ edges}$$

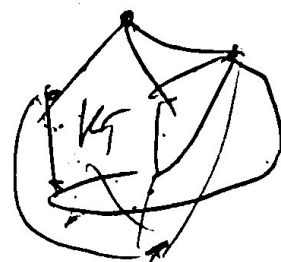
$$(n-4) 10 \text{ edges} = 10 + (n-5) \cdot 4$$

$$= 10n - 40$$

$$= 10 + 4n - 20 = 4n - 10$$

$$\sum \deg v = 2e$$

$$= 8n - 20$$



$$\frac{8n^2}{n} \approx 8 - \frac{2}{n} < 8$$

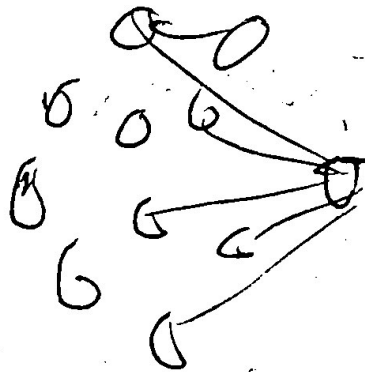
let $n=6$

$$8 - \frac{2}{6}$$

$$= 8 - \frac{1}{3}$$

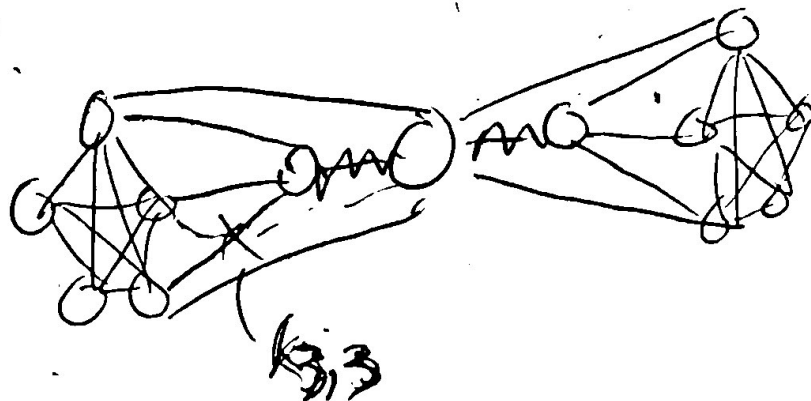
$$= 7\frac{2}{3}$$

$$\begin{matrix} R \\ G \\ B \end{matrix} \not\equiv \begin{matrix} R \\ G \\ B \end{matrix}$$

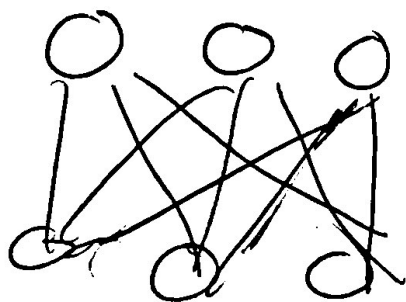


$K_{3,3}$ cycle

different

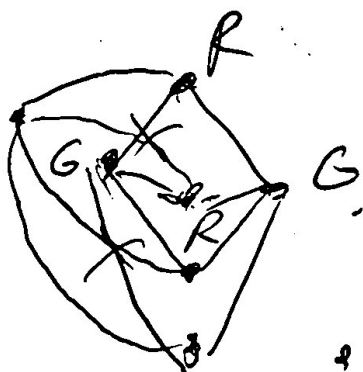
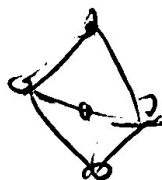
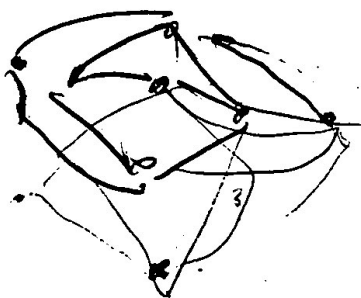


$K_{3,3}$



Next:

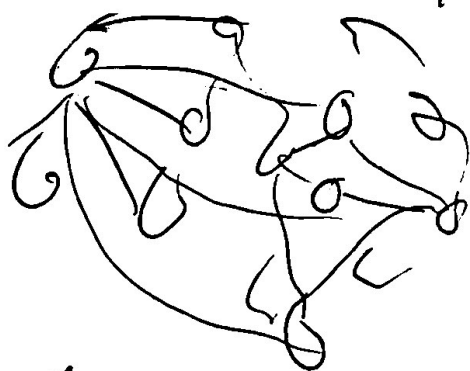
$K_{3,3}$ problem



fl node $\rightarrow + 3$ edges

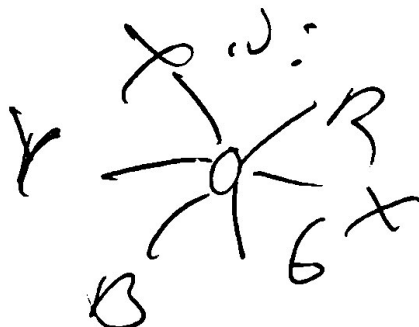
planar $K_{3,3}$

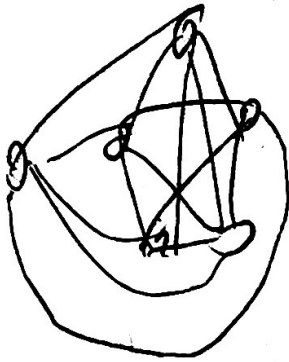
$\therefore$ every maximal $K_{3,3}$ graph
is 2-colorable



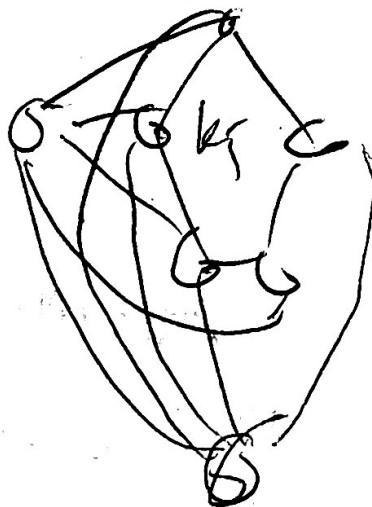
∂G

Simplex of
dual

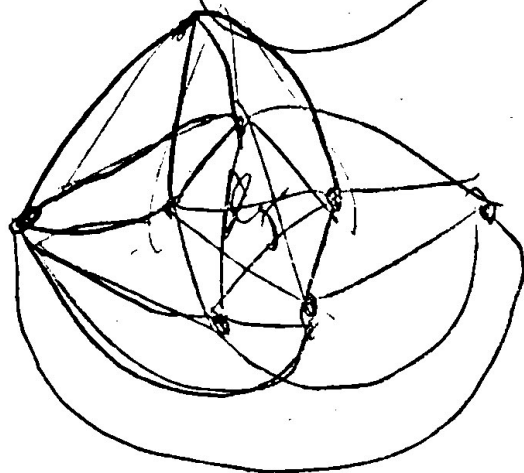
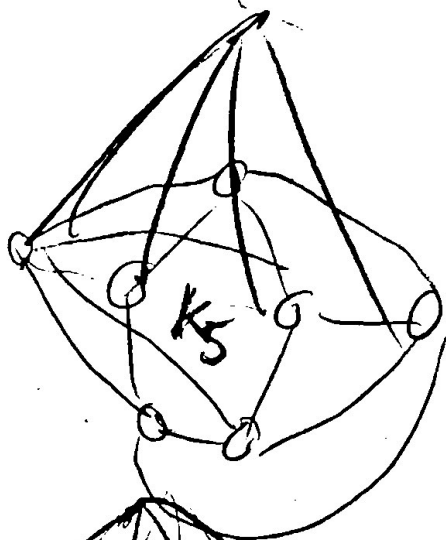




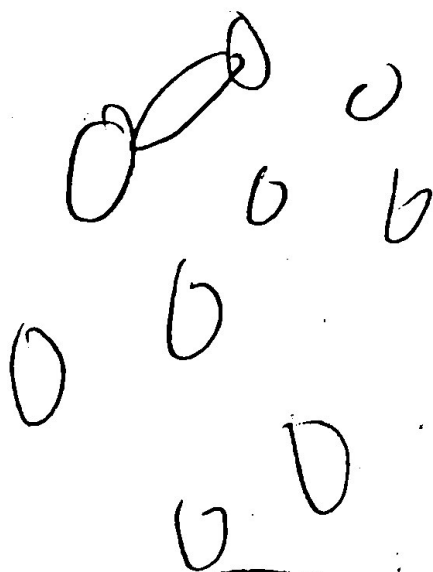
CC



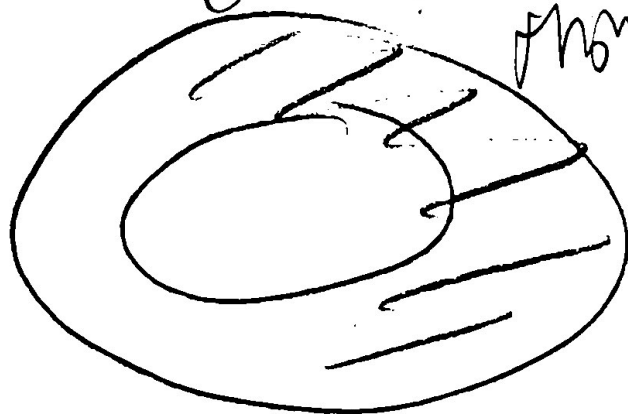
Maximall K_5 network by
every node has K_5 minor.



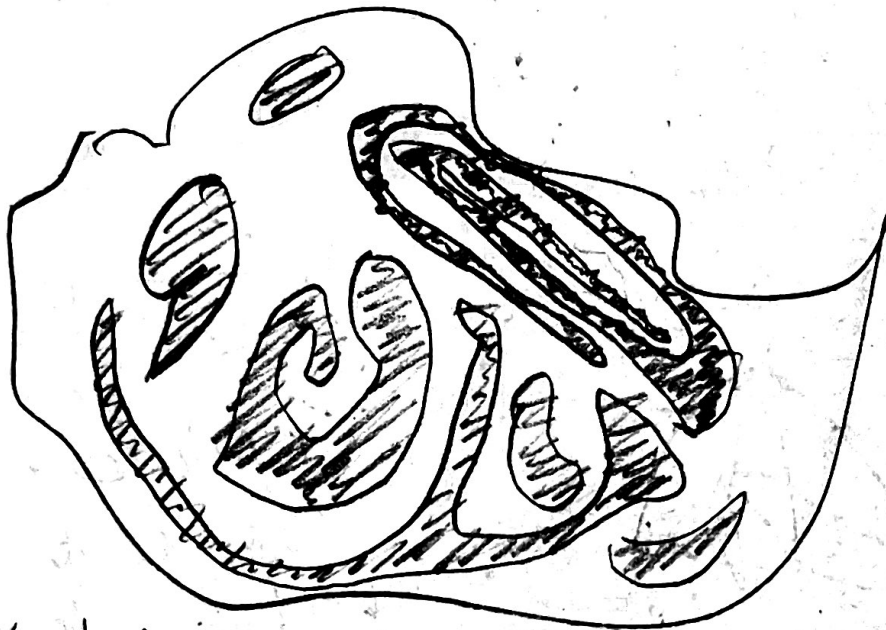
1 node can't be
connected to > 4
vertices in K_5 network
else K_6 must
rocket in



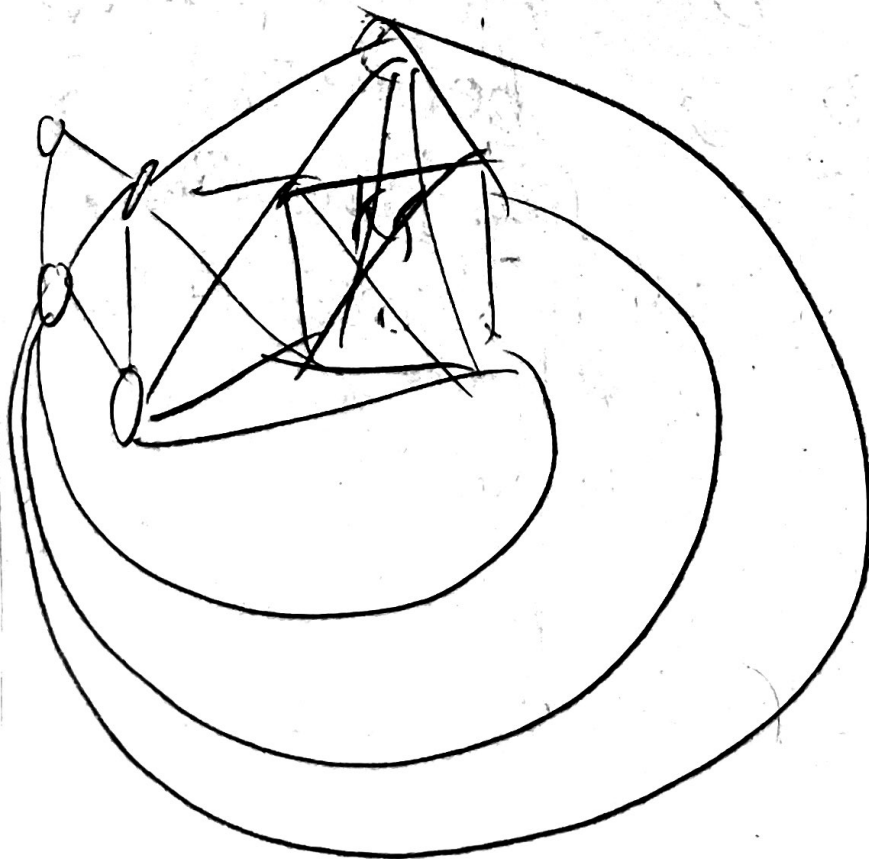
planar component with
be connected to 73
nodes in K_5 network.
If the nodes in planar
network can be connected
though!

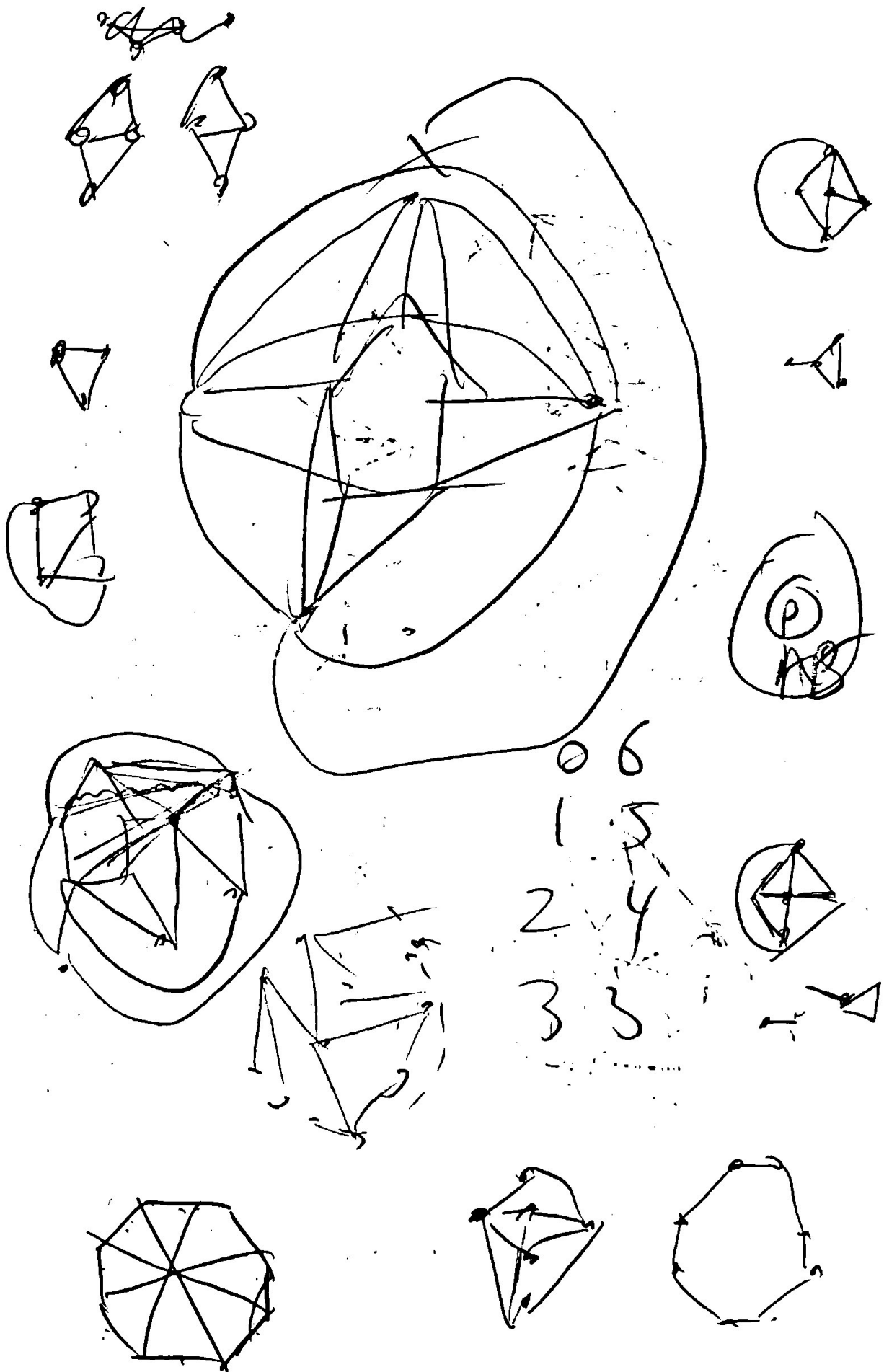


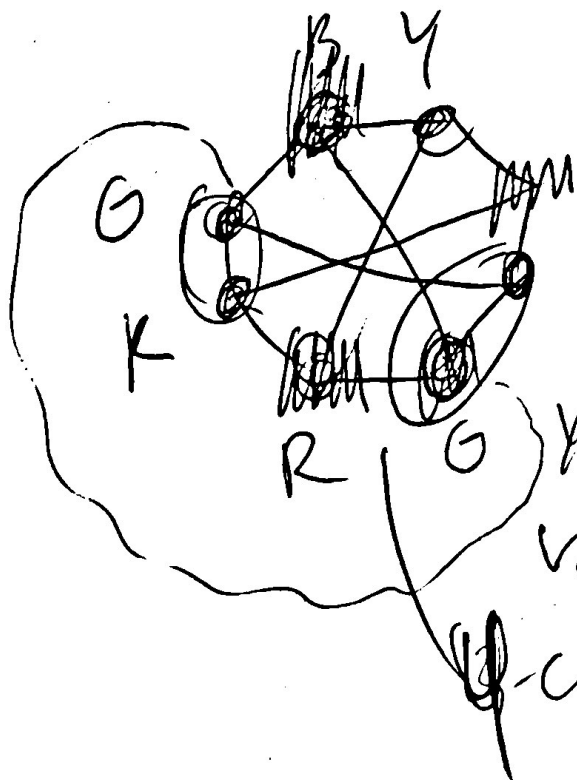
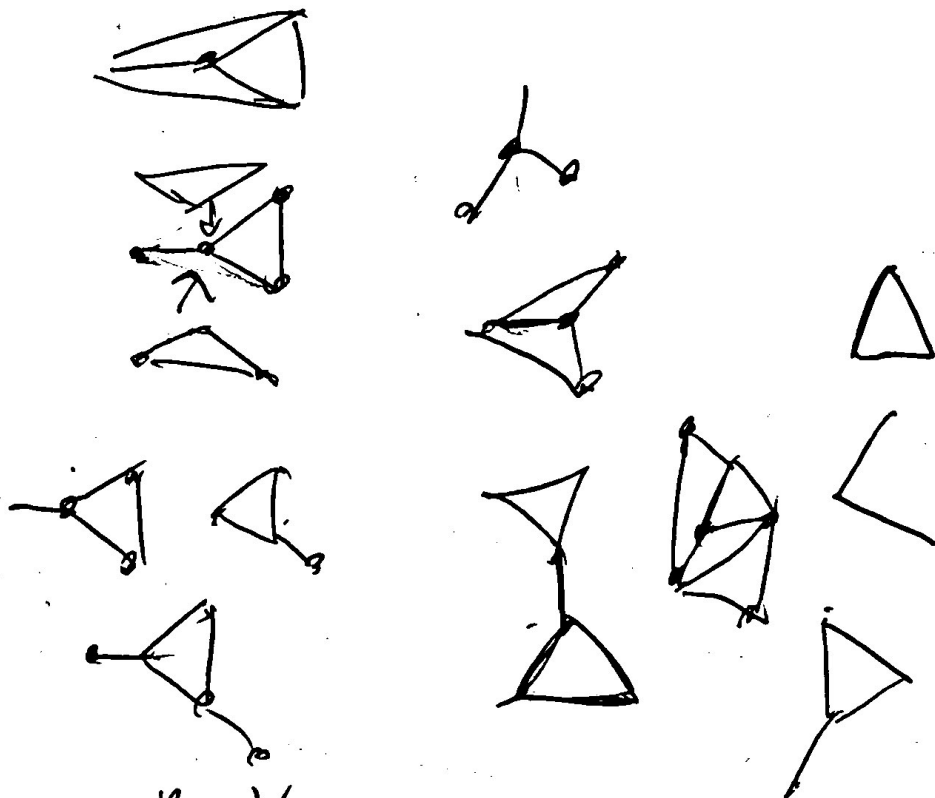
Seems no
dead block...



Shaded - planar part -
 rough - nonplanar part



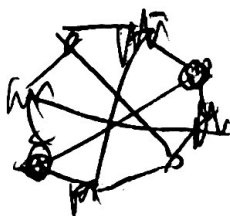
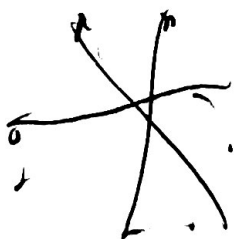
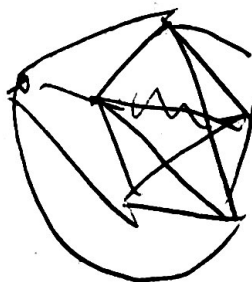
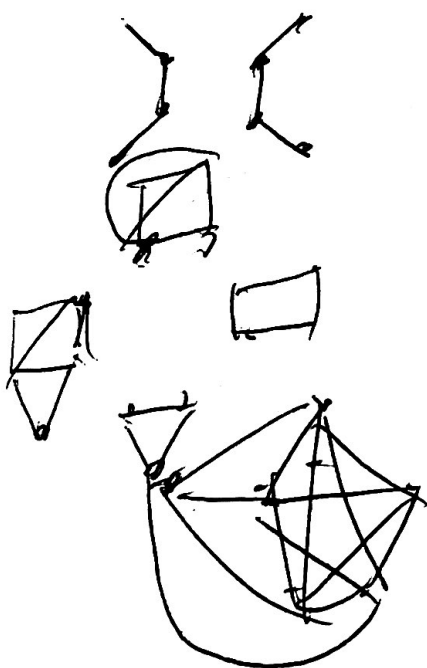
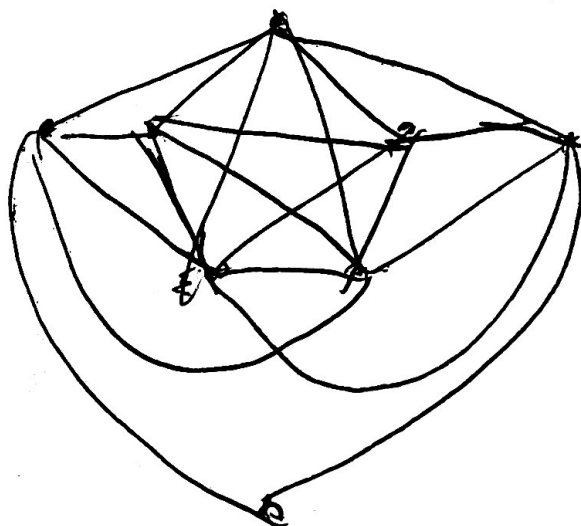
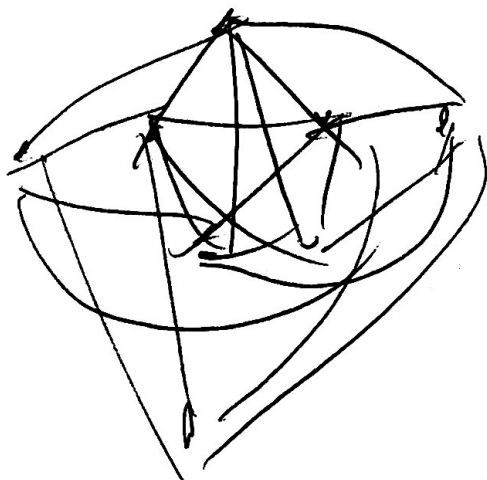


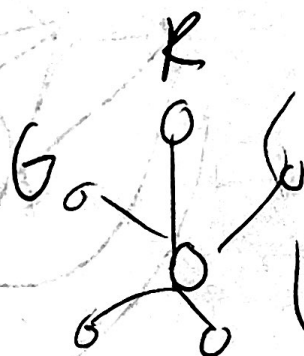
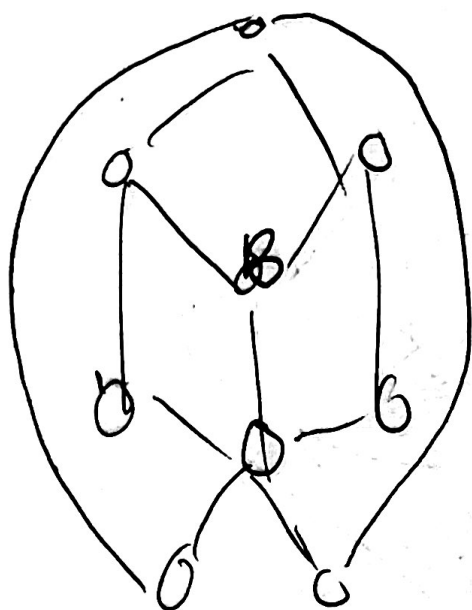


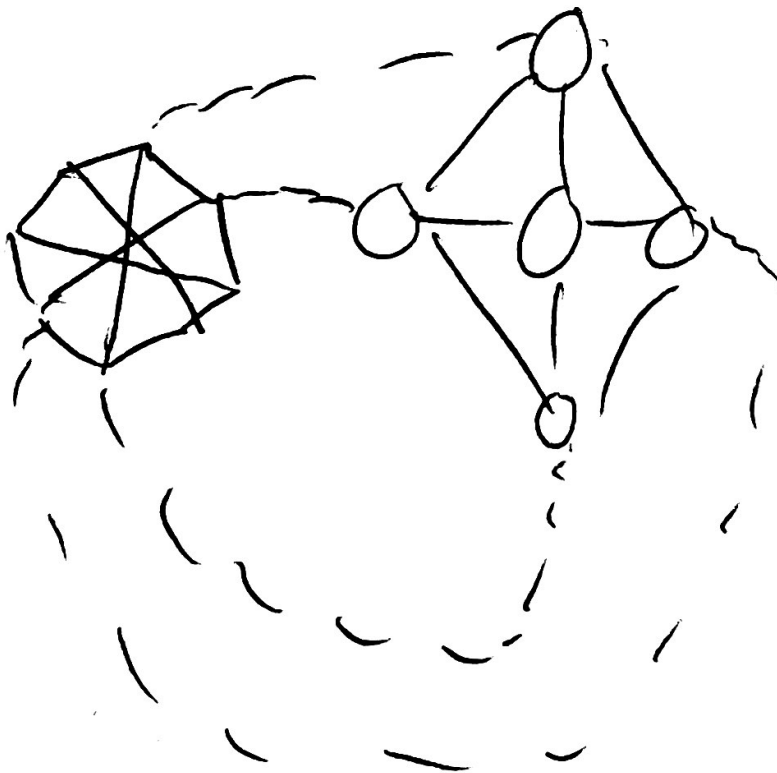
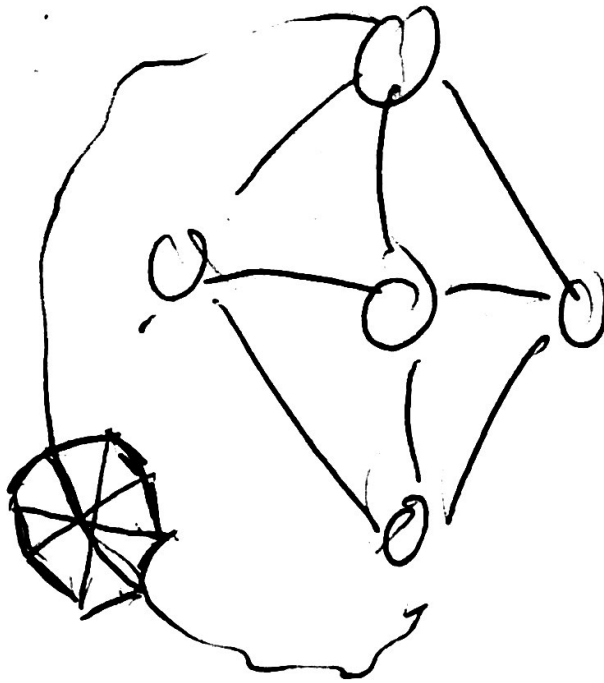
v_8 is a graph, to
any 2 cycle can
in a planar graph still
with K_3 embedded in

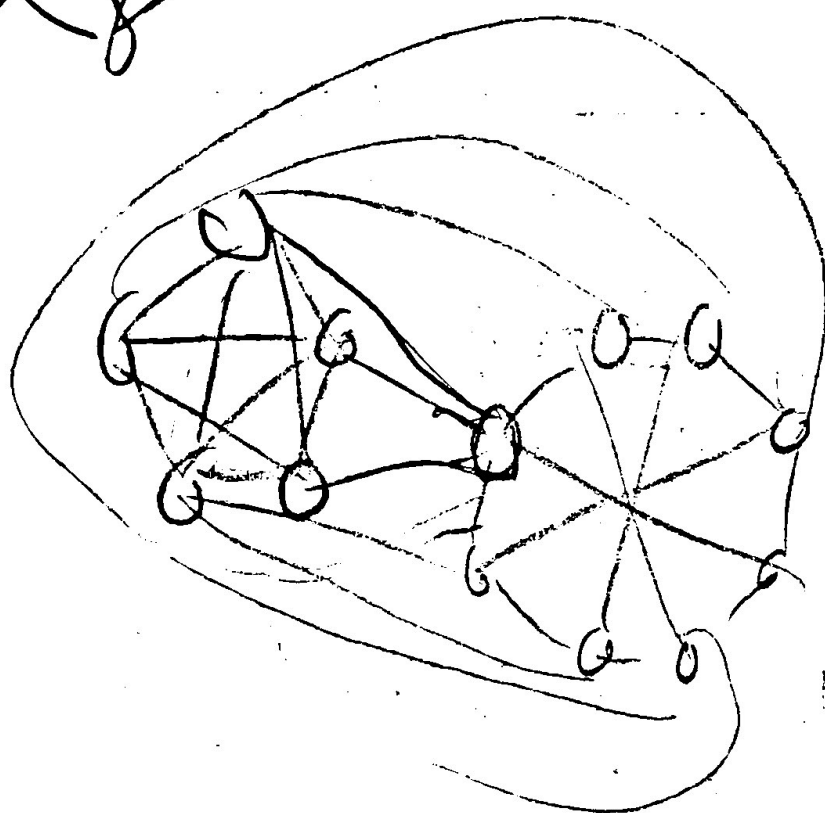
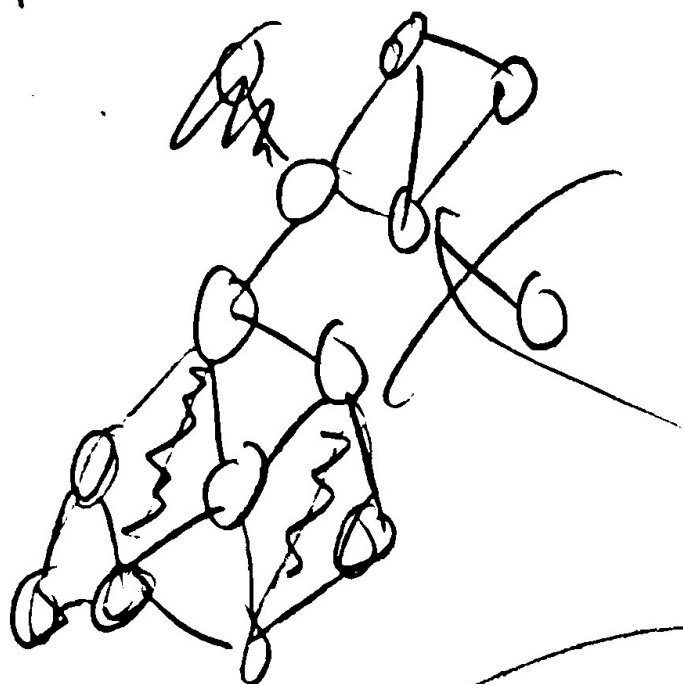
v_8
- v_8

Can contract
any 2 pairs and
still have K_3
pairs can't be area
connected







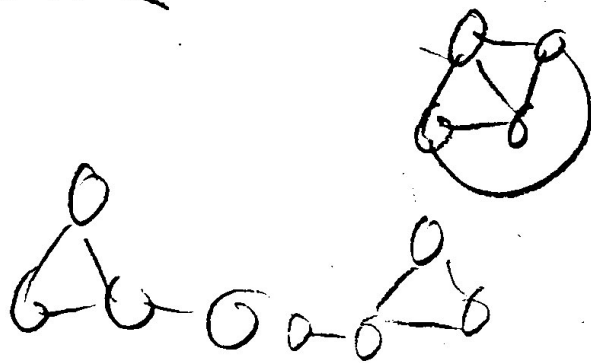


Assume G_1 and G_2 are maximally k_n graphs. $n > 0$

degeneracy (G_1, G_2)

$$= \max(\text{degeneracy}(G_1, G_2))$$

base case



G_1 and G_2 must be spanned by their minor.

- graph minor theory.

planar $\Rightarrow$ no K_5 minor

$\Rightarrow$ 5-degenerate

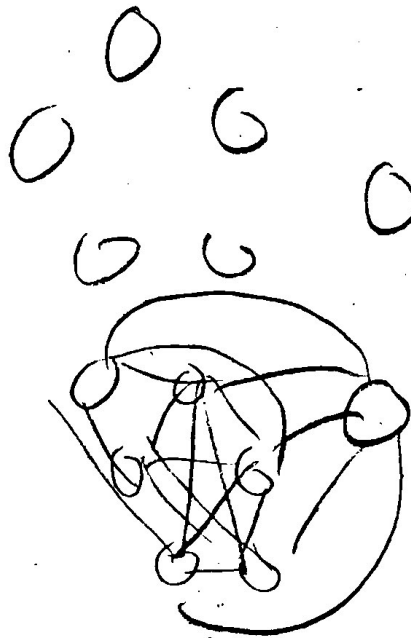
no K_6 minor $\Rightarrow$ 6-degenerate?

$$e = 4n - 10 \text{ for } K_5$$

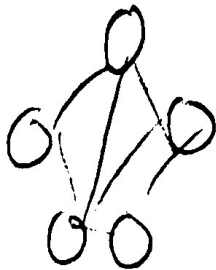
$$\Rightarrow \frac{2e}{n} = 8 - \frac{20}{n} \quad \therefore \text{not } 5\text{-degenerate, but why are maximally } K_5 \text{ graphs}$$

* graphs with degree 6 is certainly
not fulfilling of being a graph with K_6

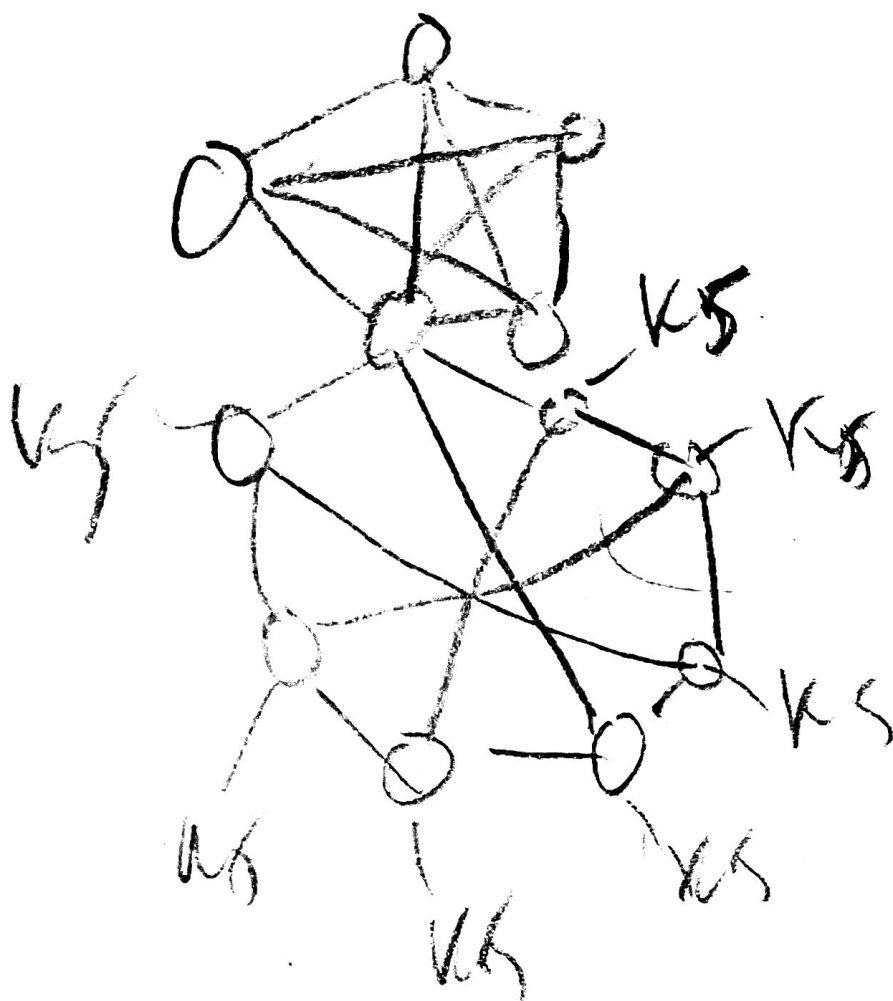
for $\text{av. degree} = 7$, $\text{min. degree (degree)} = 6$
has no K_6 minor



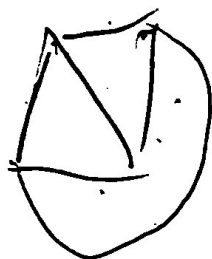
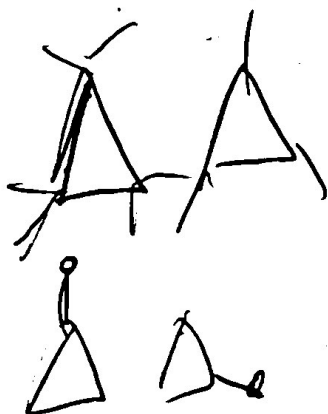
from how irregular to is a degenerate?



1/1 1-



proe degeneracy substitution



$$e = 6$$

for maximal / kg graph

$$e = 6 + 3(n-4)$$

$$= 6 + 3n - 12$$

$$e = 3n - 6$$

$$\text{avg deg.} = \frac{2e}{n} = \frac{6n-12}{n} = 6 - \frac{12}{n}$$

$\Rightarrow$ 5 degenerat

a graph w/ $n \geq 6$ mbr.

maximally k



$$e = 10$$

$$e = 10 + 4(n-5)$$

$$= 10 + 4n - 20$$

$$= 4n - 10$$

$\therefore$ 7 degenerat

a graphed as follows

$$\text{has } e \leq (t-2)v - \frac{t(t-2)}{2}$$

$$\Rightarrow 2e \leq (2t-4)v - (t^2-t)$$

$$\Rightarrow \frac{2e}{v} \leq (2t-4) - \frac{t^2-t}{v}$$

$$\Rightarrow \text{av}(\text{degree}) \leq \cancel{t^2} + \cancel{3t-4} - \frac{t^2-t}{v}$$

$$t=5$$

$$\text{av}(\text{deg}) \leq -25 + 3 \cdot 5 - 4$$

$$e \leq (t-2)v - \frac{(t-1)(t-2)}{2}$$

$$\frac{2e}{v} \leq (2t-4) - \frac{(t-1)(t-2)}{v}$$

$$= (t-1) \left(2 - \frac{t-2}{v} \right) = (t-1) \left(2 - \frac{t-2}{v} \right)$$

$$= (t-1) \left(2 - \frac{t-2}{v} \right) \text{ or } (4) \left(2 - \frac{3}{v} \right)$$

$$\text{av}(\text{deg}) \leq (t-1) \left(2 - \frac{t-2}{v} \right)$$

$$\text{an deg} \leq (t-1) \left(2 - \frac{(t-2)}{v} \right)$$

$$\text{want } \leq 2(t-1), \text{ for } t \geq 2$$

for a graph w/ no K_t minor

$$2^{t-2} = \frac{2^t}{4}$$

~~for a graph w/ K_t minor
an deg $\geq 2(t-1)$~~

$$\text{let } v = t-1$$

to get best bound on an deg

$$\text{an deg} \leq (t-1) \left(2 - \frac{(t-2)}{(t-1)} \right)$$

$$= 2t-2 - (t-2)$$

$$= t$$

no K_2 minor
 $\Rightarrow \text{andeg} \leq 2$ andeg ≥ 0

no K_6 minor
 $\Rightarrow \text{andeg} \leq (6-1)(2-(6-2))$

$$2t - \underbrace{(t^2 - 2t)}_v - 2 + \underbrace{(t-2)}_v$$

$$= 2t - 2 - \underbrace{(t-2)}_v \left(\underline{t-1} \right)$$

$$\text{andeg} \leq (t-2) \left(2 - \frac{t-1}{v} \right)$$

andeg $= t-2$ for

$$\text{let } v=1 \Rightarrow (t-2)(2 - (t-1))$$

$$= (t-2)(2-t+1)$$

$$= (t-2)(-t+3)$$

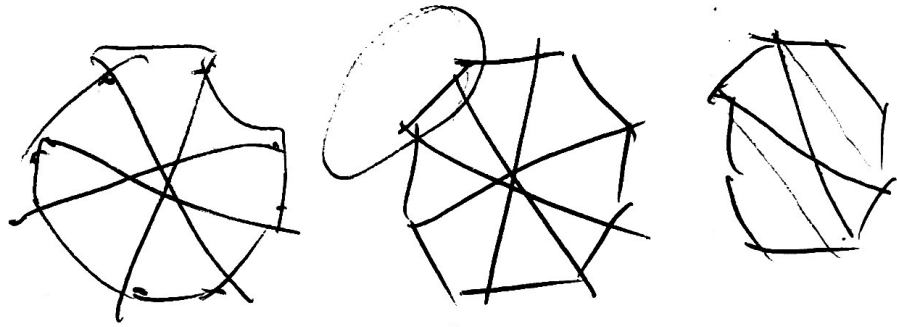
andeg $\geq$

$$t=5 \Rightarrow \text{andeg} \leq (3)(2 - \frac{4}{1})$$

andeg $\leq 2t-4$ for first graph

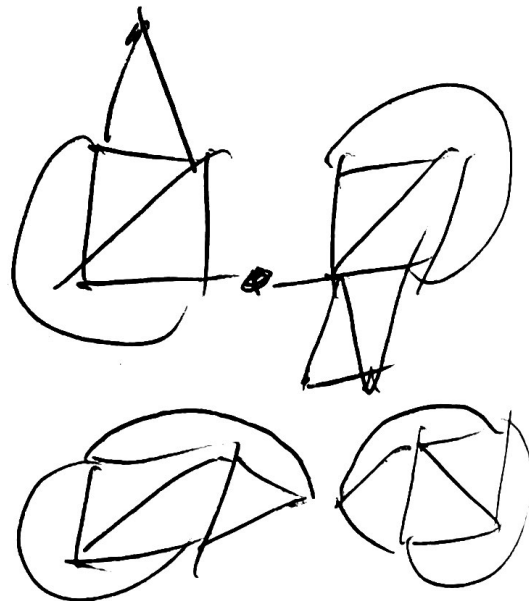
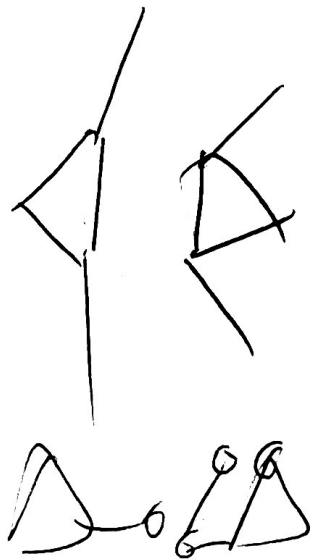
$$2(t-2) \leq 2t^2 \quad \forall t \geq 0$$

$\therefore$ minor free



K_2, K_4 minor, but K_3

K_3 subgraph

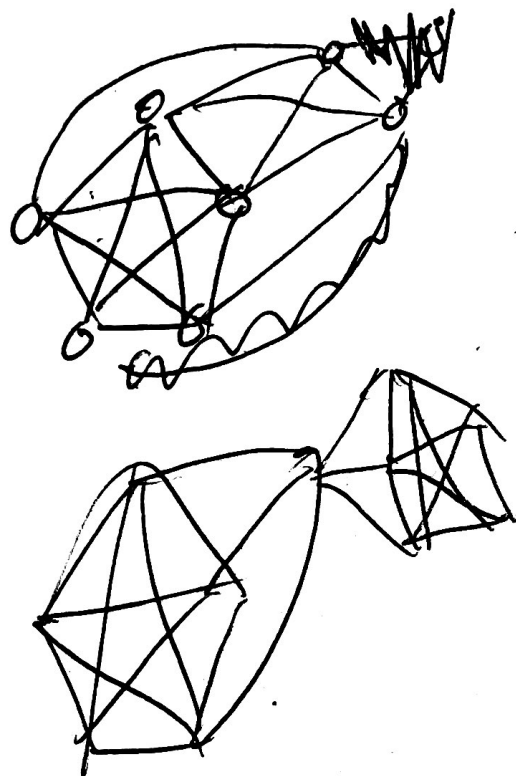


$$\text{Degeneracy}(G) = \max(\text{degen}(G_1), \text{degen}(G_2))$$

~~proof~~ Degeneracy: a k-degenerate graph:
 is an undirected graph in which every
~~induced~~ subgraph has at least one vertex of
 degree at most k .

- ~~proof~~
- ① every subgraph with a node not in clique
 sum will be $\max(\text{degen}(G_1), \text{degen}(G_2))$,
 - ② the subgraph with nodes only in clique
 sum has degeneracy $(k-1)$.

∴ No K_k minor is k -degenerate



$K_{2,2,2,2}$

6 regular